

CHARLOTTESVILLE-ALBEMARLE, VA, USA

WMO: 724016

Lat: 38.137N

Lon: 78.455W

Elev: 195

StdP: 99.00

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 93736

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-7.1	-20.6	0.6	-5.4	-17.8	0.8	-3.4	8.4	7.0	7.8	6.7	1.7	310	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	34.0	23.6	32.6	23.2	31.3	22.7	25.1	31.6	24.5	30.2	23.9	29.1	2.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.5	27.5	22.7	17.8	26.9	22.3	17.4	26.4	77.3	31.6	74.6	30.4	72.1	29.2	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.5	5.7	-13.7	35.9	2.5	2.1	-15.4	37.4	-16.9	38.6	-18.3	39.8	-20.0	41.4
			-14.4	26.5	2.3	1.0	-16.1	27.3	-17.4	27.9	-18.7	28.5	-20.4	29.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	2.5	4.2	8.3	14.0	18.6	23.0	24.9	24.1	20.5	14.5	8.9	4.6
	DBStd	9.03	5.75	5.38	5.43	4.81	4.33	3.07	2.94	2.66	3.69	4.57	4.76	4.98
	HDD10.0	801	241	174	99	16	1	0	0	0	0	12	76	181
	HDD18.3	2266	490	395	311	146	51	4	1	1	21	137	284	427
	CDD10.0	2282	9	13	48	136	268	388	462	438	313	150	43	13
	CDD18.3	706	0	0	2	16	61	142	204	180	84	16	1	0
	CDH23.3	6362	1	3	26	201	549	1263	1952	1561	653	143	9	1
	CDH26.7	2374	0	0	2	51	165	458	843	607	213	34	1	0
Wind	WSAvg	2.0	2.4	2.4	2.6	2.6	2.0	1.8	1.6	1.5	1.6	1.8	2.0	2.0
Precipitation	PrecAvg	1080	77	62	97	76	104	100	111	84	114	93	80	86
	PrecMax	1728	167	185	212	127	222	179	255	153	247	231	167	189
	PrecMin	794	27	12	5	27	27	18	30	22	14	0	9	47
	PrecStd	227	42	43	45	27	56	45	60	30	75	67	47	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.5	22.8	26.5	30.7	32.4	34.8	36.9	35.0	34.0	30.1	24.5	21.5
		MCWB	14.3	15.2	16.2	18.3	22.3	23.7	24.6	23.7	22.8	21.6	16.8	14.7
	2%	DB	17.2	18.8	23.2	28.0	30.3	32.5	34.5	33.2	31.2	27.0	21.4	17.8
		MCWB	12.5	11.9	14.4	17.3	21.3	22.8	23.9	23.5	22.3	19.2	14.2	13.5
	5%	DB	14.4	16.1	20.5	25.9	28.6	31.2	32.8	32.0	29.0	24.5	19.0	15.2
		MCWB	10.2	9.5	12.7	16.2	20.3	22.5	23.6	23.3	21.2	18.3	13.5	11.5
	10%	DB	11.6	12.8	17.8	23.0	26.7	29.4	31.2	30.2	27.2	22.4	17.2	12.5
		MCWB	7.5	7.7	11.1	14.9	19.3	21.7	23.1	22.5	20.3	17.1	12.6	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	16.5	17.5	20.6	23.8	25.1	26.0	25.9	24.6	23.1	18.4	17.5
		MCDB	18.3	20.9	22.8	26.0	29.6	31.4	33.0	32.0	30.6	27.7	21.6	19.6
	2%	WB	13.7	13.1	15.8	18.9	22.5	24.2	25.3	25.0	23.6	21.5	16.8	14.5
		MCDB	16.6	17.3	21.4	25.5	28.3	30.1	32.1	30.9	28.5	25.0	19.9	17.2
	5%	WB	10.7	10.6	14.1	17.4	21.5	23.6	24.6	24.2	22.7	19.9	14.8	11.7
		MCDB	13.5	14.4	19.0	23.5	26.8	29.0	30.7	29.2	26.7	23.1	18.2	14.4
	10%	WB	7.8	8.0	12.1	16.1	20.5	22.8	23.9	23.5	21.8	17.9	13.0	8.9
		MCDB	11.1	12.6	16.8	21.6	25.3	27.7	29.4	28.1	25.3	21.0	16.0	12.1
Mean Daily Temperature Range	5% DB	MDBR	9.3	10.4	11.4	12.2	11.4	11.0	10.8	10.7	10.9	11.5	11.1	9.4
		MCDBR	13.0	14.8	14.8	15.9	13.6	12.8	12.6	12.6	13.3	14.1	14.1	12.4
		MCWBR	9.5	9.6	8.4	7.4	6.2	5.2	4.5	4.7	5.6	7.2	8.6	8.8
	5% WB	MCDBR	11.5	12.9	13.1	13.1	11.7	11.3	11.0	10.7	10.2	10.7	10.9	10.8
		MCWBR	9.4	9.5	8.3	7.1	5.8	5.1	4.4	4.7	4.9	6.3	8.1	8.9
Clear-Sky Solar Irradiance	taub	0.297	0.315	0.349	0.393	0.437	0.484	0.505	0.512	0.408	0.354	0.319	0.300	
	taud	2.509	2.442	2.396	2.299	2.196	2.104	2.071	2.027	2.307	2.453	2.509	2.531	
	Ebn at Noon	892	915	911	886	849	806	785	768	838	858	856	863	
	Edh at Noon	78	94	109	128	145	159	163	166	119	93	78	71	
All-Sky Solar Radiation	RadAvg	2.23	3.03	4.02	5.02	5.57	6.13	5.85	5.27	4.37	3.41	2.51	1.96	
	RadStd	0.18	0.34	0.24	0.39	0.48	0.40	0.34	0.26	0.51	0.39	0.19	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (55 neighbors)	+1.02	N/A	N/A	N/A	+0.40	N/A	N/A	N/A	+322	+218
		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+127	+73

Nomenclature: See separate page